Sheet (4)

(1) Write a Python program that prompts the user to enter 10 numbers, calculates their average, and displays the result.

(2) Write a Python program that prompts the user to enter a non-negative integer, calculates its factorial, and displays the result.

(3) Write a Python program that repeatedly prompts the user to enter a positive or negative number until zero is entered. The program should then calculate and display the sum of all positive numbers and the sum of all negative numbers entered.

(4) Write a Python program that prompts the user to enter two numbers and uses a while loop to calculate and display the sums of the odd and even numbers separately within the specified range, including both endpoints.

(5) Write a Python program to find the largest n such that $2^n < 30000$. Use a while loop to find the largest integer n such that 2^n is less than 30.000.

(6) Write a Python program that prompts the user to enter an integer, calculates the sum of its digits, and displays the result.

(7) Write a Python program that prompts the user to enter an integer, calculates the product of its digits, and displays the result.

(8) Write a Python program that prompts the user to enter an integer and displays the number with its digits reversed.

(9) Write a Python program to calculate and print the sum of the first n terms of the following series

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ \ldots\ +\frac{1}{n!}$$

(10) Write a Python program to calculate the product of all terms of the following series:

$$\frac{1}{5}\ ,\ \frac{1}{9},\ \frac{1}{13}\ ,\ \frac{1}{17}\ ,\ \frac{1}{21}\ ,\ \frac{1}{25}\ ,\ \cdots\ ,\ \frac{1}{93}\ ,\ \frac{1}{97}$$

(11) The ratio π can be calculated from *Leonhard Euler's* formula which is expressed as follows; $\frac{\pi^2}{6}=\sum_{n=1}^{\infty}\frac{1}{n^2}$. Therefore π is expressed as

$$\pi = \sqrt{6 \times \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \ldots + \frac{1}{n^2}\right)}$$

Write a Python program to calculate an approximate values of π and displays these values for $n = 100.200. \quad ... \quad and\ 600.$

(12) The exponential function is expressed by *Taylor* series as $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ which can be rewritten as; $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots + \frac{x^n}{n!}$

Write a Python program that prompts the user to enter x and then calculate and displays the value of e^x for $n = 15$.

(13) Write a Python program that prompts the user to enter an integer and determines whether the number is prime or not.

(14) Write a Python program that prompts the user to enter two integers, calculates their greatest common divisor (GCD), and displays the result.

(15) Suppose a pentagon is centered at (0.0) with one point at the 0 o'clock position, as shown in the below figure. Write a Python program that prompts the user to enter the radius of the bounding circle of a pentagon and displays the coordinates of the five corner points on the pentagon.

